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PRACTICAL 3

SQL Aggregate Functions in SQL*Plus (Oracle) & MySQL

Aggregate functions perform calculations on multiple rows and return a **single** value. These functions are useful in **summarizing** data, such as totals, averages, counts, etc.

Below are detailed examples covering all aggregate functions in **SQL*Plus (Oracle)** and **MySQL**.

1. Aggregate Functions in SQL*Plus (Oracle)

1.1 SUM() – Total of a Column

```
SELECT SUM(salary) AS total_salary FROM employees; -- Total salary of
all employees
SELECT department_id, SUM(salary) FROM employees GROUP BY
department_id; -- Total salary per department
DONE
```

1.2 AVG() – Average of a Column

```
SELECT AVG(salary) AS avg_salary FROM employees; -- Average salary
SELECT department_id, AVG(salary) FROM employees GROUP BY
department_id; -- Average salary per department -DONE
```

1.3 COUNT() – Counting Rows

```
SELECT COUNT(*) FROM employees; -- Total number of employees
SELECT COUNT(employee_id) FROM employees WHERE department_id = 10; -- Count
employees in department 10
SELECT department_id, COUNT(*) FROM employees GROUP BY department_id;
-- Count per department -DONE
```

1.4 MAX() – Maximum Value

```
SELECT MAX(salary) FROM employees; -- Highest salary in the company
SELECT department_id, MAX(salary) FROM employees GROUP BY
department_id; -- Highest salary per department -DONE
```

1.5 MIN() – Minimum Value

```
SELECT MIN(salary) FROM employees; -- Lowest salary in the company
SELECT department_id, MIN(salary) FROM employees GROUP BY
department_id; -- Lowest salary per department -DONE
```

1.6 MEDIAN() – Median Value (Oracle-Only)

```
SELECT MEDIAN(salary) FROM employees; -- Median salary of all
employees
```

```
SQL> SELECT MEDIAN(salary) FROM company;

MEDIAN(SALARY)
-----
          2750000
```

1.7 STDDEV() – Standard Deviation

```
SELECT STDDEV(salary) FROM employees; -- Standard deviation of
salaries
SELECT department_id, STDDEV(salary) FROM employees GROUP BY
department_id; -- Std deviation per department
```

```
SQL> SELECT empid, STDDEV(salary) FROM company GROUP BY empid;

  EMPID  STDDEV(SALARY)
-----
      1              0
      2              0
      4              0
      3              0
```

1.8 VARIANCE() – Variance of a Column

```
SELECT VARIANCE(salary) FROM employees; -- Variance of salaries
```

```
SQL> SELECT VARIANCE(salary) FROM company;
```

```
VARIANCE(SALARY)
```

```
-----  
4.1667E+11
```

1.9 GROUP BY with Aggregate Functions

```
SELECT department_id, COUNT(*), AVG(salary), MAX(salary), MIN(salary)  
FROM employees
```

```
GROUP BY department_id; -- Aggregate calculations per department
```

```
SQL> SELECT empid, COUNT(*), AVG(salary), MAX(salary), MIN(salary) FROM company  
2 group by empid;
```

EMPID	COUNT(*)	AVG(SALARY)	MAX(SALARY)	MIN(SALARY)
1	1	2000000	2000000	2000000
2	1	3000000	3000000	3000000
4	1	3500000	3500000	3500000
3	1	2500000	2500000	2500000

1.10 HAVING Clause (Filtering Groups)

```
SELECT department_id, COUNT(*) AS employee_count
```

```
FROM employees
```

```
GROUP BY department_id
```

```
HAVING COUNT(*) > 5; -- Only departments with more than 5 employees
```

```
SQL> SELECT empid, COUNT(*) AS employee_count
2  from company
3  group by empid
4  having count(*)<5;
```

EMPID	EMPLOYEE_COUNT
1	1
2	1
3	1
4	1

2. Aggregate Functions in MySQL

2.1 SUM() – Total of a Column

```
SELECT SUM(salary) AS total_salary FROM employees; -- Total salary of
all employees
```

```
SELECT department_id, SUM(salary) FROM employees GROUP BY
department_id; -- Total salary per department
```

2.2 AVG() – Average of a Column

```
SELECT AVG(salary) AS avg_salary FROM employees; -- Average salary
```

```
SELECT department_id, AVG(salary) FROM employees GROUP BY
department_id; -- Average salary per department
```

2.3 COUNT() – Counting Rows

```
SELECT COUNT(*) FROM employees; -- Total number of employees
SELECT COUNT(employee_id) FROM employees WHERE department_id = 10; -- Count
employees in department 10
```

```
SELECT department_id, COUNT(*) FROM employees GROUP BY department_id;  
-- Count per department
```

2.4 MAX() – Maximum Value

```
SELECT MAX(salary) FROM employees; -- Highest salary in the company  
SELECT department_id, MAX(salary) FROM employees GROUP BY  
department_id; -- Highest salary per department
```

2.5 MIN() – Minimum Value

```
SELECT MIN(salary) FROM employees; -- Lowest salary in the company  
SELECT department_id, MIN(salary) FROM employees GROUP BY  
department_id; -- Lowest salary per department
```

2.6 STDDEV() – Standard Deviation

```
SELECT STDDEV(salary) FROM employees; -- Standard deviation of  
salaries  
SELECT department_id, STDDEV(salary) FROM employees GROUP BY  
department_id; -- Std deviation per department
```

2.7 VARIANCE() – Variance of a Column

```
SELECT VARIANCE(salary) FROM employees; -- Variance of salaries
```

2.8 GROUP BY with Aggregate Functions

```
SELECT department_id, COUNT(*), AVG(salary), MAX(salary), MIN(salary)  
FROM employees
```

```
GROUP BY department_id; -- Aggregate calculations per department
```

2.9 HAVING Clause (Filtering Groups)

```
SELECT department_id, COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5; -- Only departments with more than 5 employees
```

3. Key Differences Between SQL*Plus (Oracle) and MySQL Aggregate Functions

Feature	Oracle (SQL*Plus)	MySQL
Basic Aggregate Functions	<code>SUM()</code> , <code>AVG()</code> , <code>COUNT()</code> , <code>MAX()</code> , <code>MIN()</code>	<code>SUM()</code> , <code>AVG()</code> , <code>COUNT()</code> , <code>MAX()</code> , <code>MIN()</code>
Median Calculation	<code>MEDIAN()</code>	Not available (requires workaround)
Standard Deviation	<code>STDDEV()</code>	<code>STDDEV()</code>
Variance Calculation	<code>VARIANCE()</code>	<code>VARIANCE()</code>
Handling NULL values	Ignores NULL values in aggregate functions	Ignores NULL values in aggregate functions
HAVING Clause	Used after <code>GROUP BY</code>	Used after <code>GROUP BY</code>
GROUP BY	Used to group and aggregate	Used to group and aggregate

4. Special Notes

- **Oracle** provides the `MEDIAN()` function, but **MySQL does not**. In MySQL, median must be calculated using a workaround.

- **Both** ignore NULL values when computing aggregates unless explicitly handled. •
- HAVING** is used **after GROUP BY** to filter aggregated results in both Oracle and MySQL.